

Joint Radar-Communication Optimization Using SEFDM Waveforms and Intelligent Element Grouping -IRS

Arnav Khond

Abstract- The 6G ISAC systems experience an essential technical challenge where the SEFDM technique improves the spectral efficiency; however, it introduces significant Inter-Carrier Interference (ICI) effects that negatively impact the radar detection performance whereas the Extremely Large Intelligent Reflecting Surface creates tremendous signals; on the other hand, it requires very high computational complexity. To overcome this issue, we suggest using the SEFDM signal formation scheme through the intelligent element grouping IRS (IEG-IRS). Here, by separately designing the SEFDM ICI effect in time-frequency domain and the cascaded spatial channel for IEG-IRS, we design a unified decoupled receiver signal model to show the interference cancellation from the "deterministic virtual LOS" of grouped IRSs. In conclusion, we verify that array gains of IEG-IRS compensate the cost of ICI penalty, optimizing joint performance for both communication and radar.

Keywords: ISAC-integrated systems and communications, SEFDM-spectrally efficient Frequency division multiplexing.

I. Introduction

The integration of sensing and communication (ISAC) is rapidly emerging as a foundational pillar for next-generation (6G) wireless networks, driven by the demand for ubiquitous connectivity and high-precision environmental awareness. By enabling a single hardware platform and waveform to simultaneously transmit data and detect targets, ISAC promises unprecedented spectral and hardware efficiency.

However, realizing this potential requires waveforms that can satisfy the conflicting demands of both functions.

The Waveform Conflict: SEFDM vs. OFDM

Traditional Orthogonal Frequency Division Multiplexing (OFDM) is widely used due to its robustness and simple transceiver architecture. However, its strict subcarrier orthogonality fundamentally limits spectral efficiency. Spectrally Efficient Frequency Division Multiplexing (SEFDM) addresses this by deliberately compressing the subcarrier spacing below the Nyquist rate. While this significantly boosts data rates, the resulting non-orthogonality introduces severe Inter-Carrier Interference (ICI).

Communication Impact: ICI makes detection of the data very difficult and requires a highly complex, expensive receiver.

Interference for the Radar Sensing: Due to this ICI, self-interference occurs in radar applications and greatly degrades the precision of range and Doppler estimation, especially when multiple targets are present leading to massive entanglement of echoes.

The Spatial Solution: Intelligent Reflecting Surfaces

On the other hand, Intelligent Reflecting Surfaces (IRS) have received huge attention for actively altering the wireless propagation environment. An IRS can effectively combat blockage and severe path loss (the so-called "multiplicative fading" phenomenon) by smartly manipulating the phase shifts of large arrays of passive reflecting elements to generate strong, directed signal beams.

Yet, the implementation of Extremely Large-scale IRS (XL-IRS) in practice is bottlenecked by two

Signalling Overhead: Tracking the high-dimensional cascaded channel involves an infeasible amount of pilot signals.

So, to derive their phase shift we have to open the pattern recognition window in real-time which would require prohibitive amounts of processing power.

Proposed ISAC Synthesis of SEFDM and IRS To leverage the advantages of SEFDM and IRS and circumvent the problems inherent in each technology, it is proposed to synthesize the SEFDM waveform together with the Intelligent Element Grouping IRS (IEG-IRS). The IEG-IRS approach pre-group the huge antenna array to several groups where the elements within each group have a common phase shift. This helps reduce the overhead as well as complexity but still offers massive antenna gain acting as a "virtual deterministic LOS."

Research Question: In order to assess the effectiveness of the synthesis of SEFDM waveforms and the IEG-IRS, how is their joint output characterized by modelling the ICI of SEFDM waveforms and cascaded channel of IEG-IRS separately?

The present research mathematically characterizes a unified, decoupled received signal model for evaluating the above synthesis. We analyze the temporal frequency ICI penalty of SEFDM and the spatially cascaded channel of IEG-IRS to investigate an ISAC trade-off.

II. System Model

In this paper, we study a downlink ISAC system that consists of a BS, which transmits data to a communication user and parameter estimation for a sensing target. To reduce significant path loss and blockage, IEG-IRS is introduced to support the communication process. To simplify the analysis of the system, we decompose the system model into two independent models: temporal frequency waveform domain and spatial cascaded channel domain.

A. SEFDM Transmitted Signal Model (Temporal-Frequency Domain)

To improve spectral efficiency, the BS transmits the data using a SEFDM waveform. Unlike conventional OFDM schemes that use orthogonal subcarriers with a fixed distance $\Delta f = 1/T$, SEFDM compresses the distance between subcarriers by a compression factor α , where $0 < \alpha \leq 1$.

Let N be the number of subcarriers in the waveform. Then, the transmitted SEFDM signal can be expressed in the following way:

Let N denote the total number of subcarriers. The discrete-time baseband transmitted SEFDM signal for a single symbol block can be formulated as:

$$x[n] = \left(\frac{1}{\sqrt{N}}\right) \sum_{k=0}^{N-1} s_k e^{\frac{j2\pi n k \alpha}{N}}, \quad \text{for } n = 0, 1, \dots, N-1$$

where:

- s_k denotes the complex data symbol (M-QAM/M-PSK) being sent over the k -th subcarrier.
- α is the bandwidth compression ratio. For $\alpha=1$, the scheme becomes the classical orthogonal OFDM scheme.
- $\frac{1}{\sqrt{N}}$ is the power normalization factor.

This bandwidth compression will make the subcarriers non-orthogonal ($0 < \alpha < 1$). The deliberate overlapping of spectrum results in the inter-carrier interference (ICI) on the receiver end, which becomes the mathematical penalty and needs to be tackled by the advantage provided by the IRS in the spatial domain.

B. Cascaded Channel Model between IEG-IRS and XL-IRS (Spatial Domain)

We assume that there is an XL-IRS consisting of M passive reflecting units. Each reflecting unit m in an ungrouped IRS (U-IRS) would need an individual phase shift $v_m = e^{j\theta_m}$. In order to avoid the impracticality of having such a huge number of units for M , we have adopted the IEG-IRS approach.

The M reflecting units are split into Q groups. Here, $Q < M$. G_q stands for the index set representing the units of the q -th

group. It is imperative that all units belonging to G_q have a same reflection coefficient, and that is

$$v_q = e^{j\{\theta\}_q}$$

Let $h_{BI} \in \mathbb{C}^{M \times 1}$ is the channel vector from BS to the IEG-IRS and $h_{IU} \in \mathbb{C}^{M \times 1}$ is the channel vector from the IEG-IRS to the user (or target). We can define the effective cascaded channel, h_{eff} created by the IEG-IRS in the following form:

$$h_{eff} = \sum_m^Q v_1 \sum [h_{IU}]_m [h_{BI}]_m$$

It should be clear from the above equation that the optimization space has been shrunk from M elements to Q clusters and thus creates a deterministic virtual Line-of-Sight (LoS) path.

C. Synthesized ISAC Received Signal

Next, we combine the independently established definitions of SEFDM waveform penalty (ICI) and IEG-IRS spatio-temporal gain for synthesis of end-to-end received signal model. Following perfect synchronization, the discrete-time received signal at the user (or equivalently, the sensing echo at the BS) is given by a product of transmitted SEFDM-Signal and effective IEG-IRS cascaded channel corrupted by noise:

$$y[n] = h_{eff} \cdot x[n] + w[n]$$

Now, replacing decoupled formulations in the synthesized model:

$$y[n] = \left\{ \left(\sum_m^Q v_q \sum [h_{IU}]_m [h_{BI}]_m \right) * \left(\left(\frac{1}{\sqrt{N}} \right) \sum_{k=0}^{N-1} s_k e^{j \frac{2\pi n k \alpha}{N}} \right) + w[n] \right\}$$

where $w[n] \sim \mathcal{CN}(0, \sigma^2)$ is the Additive White Gaussian Noise (AWGN) with variance σ^2

This composite framework separates the two antagonistic physical effects: the phase-coherent array gain due to v_q and the non-orthogonal interference defection by α . Therefore, this model will be the fundamental mathematics basis for our joint optimization of communication data rate and the radar sensing accuracy in later analysis.

```
=====
ISAC IRS BENCHMARK: GROUPED vs INDIVIDUAL
=====

--- METHOD 1: GROUPED (IEG-IRS) ---
Time Taken      : 1.144 ms
Effective Channel : 197.99 + 113.14j
Sample Rx[0]    : 678.83 + 2488.95j

--- METHOD 2: INDIVIDUAL (Standard IRS) ---
Time Taken      : 17.977 ms
Effective Channel : 0.75 + 2.75j
Sample Rx[0]    : -16.09 + 27.93j

=====
SPEEDUP FACTOR: Grouped method is 15.7x faster!
=====
```

```
Input 1 completed in: 1.683 milliseconds (Effective Channel: 102.65 + -109.91j)
Input 25 completed in: 0.374 milliseconds (Effective Channel: -78.18 + -81.68j)
Input 50 completed in: 0.268 milliseconds (Effective Channel: -107.53 + 110.57j)
Input 75 completed in: 0.180 milliseconds (Effective Channel: 122.40 + -16.86j)
Input 100 completed in: 0.148 milliseconds (Effective Channel: -76.64 + 88.52j)

=====
TEST COMPLETE: EFFICIENCY REPORT
=====

Total inputs processed : 100
Total processing time : 34.535 ms
Average time per input : 0.345 ms

Theoretical Insight:
By grouping 1000 IRS elements into just 4 groups,
the computational complexity scaled with O(N * Q) instead of O(N^2).
This is why the average processing time remained strictly under a few milliseconds.
```

Deployment and simulation: The two approaches 6G IRS grouping simplifies the math in somehow a very faster way, while SEFDM keeps the radio signal condensed.

III. Problem Formulation

We then derive a joint optimization problem as validation of the proposed ISAC system, which maximizes the communication performance subject to a minimum required accuracy level for radar sensing.

B. Optimization Problem

We aim to optimize the IRS group phase shifts (v) and SEFDM compression factor α in order to maximize the sum-rate while considering the sensing, power, and hardware constraints. Therefore, the problem formulation can be stated as follows:

A. Communication and Sensing Metrics

The sum-rate R is the most relevant metric for the communication link, which is highly dependent on the SINR.

Because of the SEFDM waveform, the SINR on the k -th subcarrier incorporates the ICI cost in addition to AWGN:

$$R = \sum_{k=0}^{N-1} (\log_2(1 + \text{SINR}_k(\alpha, \mathbf{v})))$$

Here, $\mathbf{v}$ denotes the phase shift vector of IRS, which contains all groups, as shown below. That is, $\mathbf{v} = [\underline{V}_1, \underline{V}_2, \dots, \underline{V}_Q]^T$. Also, α refers to the bandwidth compression factor.

For the sensing operation of radar, we will use CRLB or sensing beam pattern gain as our performance measure. We define $P_s(\mathbf{v}, \alpha)$ as the sensing power aimed at the target.

B. Joint Optimization Problem

Our goal is to compute the IRS group phase shift ($\mathbf{v}$) and SEFDM compression factor α which give the highest data rate under the constraints on sensing, energy expenditure, and hardware limitations. In mathematical terms, it means:

$$\begin{aligned} & \max_{\{\mathbf{v}, \alpha\}} R(\alpha, \mathbf{v}) \\ & \text{s. t. } P_s(\mathbf{v}, \alpha) \geq \gamma_{\text{radar}} \\ & |\underline{V}_q| = 1, \quad \forall q \in \{1, 2, \dots, Q\} \\ & 0 < \alpha \leq 1 \end{aligned}$$

where γ_{radar} is the minimum required sensing threshold, and $|\underline{V}_q| = 1$, guarantees the reflecting characteristics of the IRS units.

IV. Proposed Optimization Algorithm

Given the complex nature of the optimization problem with a non-convex objective function, coupled with α , the mathematical approach to solving it is difficult. Thus, we propose an AO method with low computational requirements:

Fixing α to optimize $\mathbf{v}$: We fix the value of the SEFDM compression factor and optimize the Q IRS group phase shifts with the help of SCA or Manifold Optimization. Since here we optimize only the

Q IRS groups and not M individual elements (with $Q \ll M$), there will be a huge saving in terms of computation.

Fixing $\mathbf{v}$ to optimize α : allowing the maximization of the virtual LoS by the IRS array, α is optimized such that spectral compression is maximized to the point that ICI overwhelms spatial gain in the system.

IV. Simulation and Expected Results

The proposed scheme is tested using a simulation in ISAC using established 6G channel models. The two baseline schemes are:

- **Baseline 1:** Conventional OFDM with fully ungrouped IRS (U-IRS).
- **Baseline 2:** Conventional SEFDM without IRS assist. In our simulations, we pay attention to three Key Performance Indicators (KPIs):

Radar-Communication Trade-off:

Using different levels of radar, we obtain the Pareto front for the system. In the expected spatial gains of IEG-IRS can efficiently mitigate the ICI of SEFDM, ensuring a good data rate while still being able to detect radar targets.

Spectral Efficiency vs. Signal Compression: We shall plot the sum-rate as a function of α , finding the sweet spot where the spectral efficiency peaks before the ICI penalty effect.

Complexity Analysis: A comparison of run-time of algorithms in optimizing M elements vs Q groups of elements, demonstrating the viability of such system architecture.

V. Conclusion

The current work presents a breakthrough approach that can overcome a key constraint in future generation 6G ISAC through the synthesis of SEFDM waveform in conjunction with an Intelligent Element Grouping IRS. It was shown how through mathematical formulation of the time-frequency interference and spatial cascading channel, the large array benefits associated with the grouped IRS

elements could be used to counteract the inherent ICI drawback of SEFDM signals. In addition, it was further discussed that using a partitioned XL-IRS

with similar phase shift among the elements reduces the need for costly signaling and computation.

Main References

- [1] M. Mirabella, P. Di Viesti, C. Masouros, and G. M. Vitetta, "Joint range and Doppler estimation using spectrally efficient FDM," *IEEE Transactions on Wireless Communications*, vol. 25, pp. 11113–11127, 2026. <http://ieeexplore.ieee.org/document/11372622>
- [2] S. Zhang, T. Ji, M. Hua, Y. Huang, and L. Yang, "Element-grouping strategy for intelligent reflecting surface: Performance analysis and algorithm optimization," *IEEE Transactions on Wireless Communications*, pp. 1–1, 2026. <https://ieeexplore.ieee.org/document/11396439>
- [3] Michele Mirabella, Member, IEEE, Pasquale Di Viesti, Member, IEEE, Christos Masouros, Fellow, IEEE and Giorgio M. Vitetta[‡], Senior Member, IEEE "Joint radar-communication optimization using SEFDM waveforms and intelligent element grouping-IRS," *Zenodo*, 2026. [Online]. Available: <https://zenodo.org/records/17941153>

Additional Recommended References

- [4] F. Liu, C. Masouros, A. P. Petropulu, H. Griffiths, and L. Hanzo, "Joint radar and communication design: Applications, state-of-the-art, and the road ahead," *IEEE Transactions on Communications*, vol. 68, no. 6, pp. 3834–3862, Jun. 2020.
- [5] Q. Wu and R. Zhang, "Towards smart and reconfigurable environment: Intelligent reflecting surface aided wireless network," *IEEE Communications Magazine*, vol. 58, no. 1, pp. 106–112, Jan. 2020.
- [6] T. Xu and I. Darwazeh, "Spectrally efficient FDM: Spectrum saving technique for 5G and beyond," in *Proc. IEEE Int. Symp. Personal, Indoor and Mobile Radio Commun. (PIMRC)*, Montreal, QC, Canada, 2017, pp. 1–5.